

Daily Server Backup Procedure

As a Linux Administrator, implementing a reliable daily backup strategy for servers is crucial for data integrity and disaster recovery. Below are the detailed steps and commands for automating daily backups using shell scripts and cron jobs.

1. Backup Requirements

- Backup critical directories: `/etc`, `/var/www`, `/home`, and `/var/lib/mysql`
- Store backups in `/backup/daily`
- Compress backups to save space
- Automate the backup using `cron`

2. Create Backup Directory

```
sudo mkdir -p /backup/daily
sudo chown root:root /backup/daily
sudo chmod 700 /backup/daily
```

Explanation:

- `mkdir -p /backup/daily`: Creates the backup directory.
- `chown root:root`: Sets ownership to root.
- `chmod 700`: Restricts access to root only.

3. Backup Script

Create a script `/usr/local/bin/daily_backup.sh`:

```
#!/bin/bash

# Variables
BACKUP_DIR="/backup/daily"
DATE=$(date +"%Y-%m-%d")

# Directories to backup
DIRS="/etc /var/www /home /var/lib/mysql"

# Create a compressed backup
```

```
tar -czf "$BACKUP_DIR/server_backup_$(date +%Y%m%d).tar.gz" $DIRS  
  
# Delete backups older than 7 days  
find "$BACKUP_DIR" -type f -name "*.tar.gz" -mtime +7 -exec rm -f {} \;
```

Explanation:

- `tar -czf`: Compresses and archives specified directories.
- `find ... -mtime +7`: Deletes backups older than 7 days.

4. Make the Script Executable

```
sudo chmod +x /usr/local/bin/daily_backup.sh
```

Explanation:

- `chmod +x`: Makes the script executable.

5. Automate with Cron Job

Edit the cron jobs:

```
sudo crontab -e
```

Add the following line:

```
0 2 * * * /usr/local/bin/daily_backup.sh
```

Explanation:

- Runs the backup script daily at 2:00 AM.

6. Verify Backup

List backup files:

```
ls -lh /backup/daily
```

Test a backup file:

```
tar -tzf /backup/daily/server_backup_$(date +"%Y-%m-%d").tar.gz
```

7. Optional: Remote Backup with rsync

```
rsync -avz /backup/daily/ user@remote_server:/remote/backup/daily/
```

Explanation:

- Syncs local backups to a remote server for additional redundancy.

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